

Video 7.1.1

A brief introduction to *Induction*.

We are about to study **Determinants**.

The proofs in this topic will depend heavily on a proof technique called *Induction*.

Yes, you'll be expected to be able to write an induction proof in this course. You'll find Induction is a straightforward idea when try it.

Because some students haven't met this topic before we will give a brief introduction of how induction proofs are organised. The proofs in this topic will give you a lot more specific examples.

What sort of things does induction prove?

Induction is useful for proving a sequence of facts like the following:

(Fact #1), (Fact #2), (Fact #3), ..., (Fact #n), ...

Instead of writing "*Fact #n*" we sometimes write P_n which means Proposition number n .

A simple example:

P_n is the statement that $1 + \dots + n = \frac{n}{2}(n+1)$. $\underbrace{1+2+3+4 = \frac{4}{2}(4+1)}_{P_4}$

P_n is the statement “ $1 + 2 + \dots + n$ equals $\frac{n}{2}(n + 1)$ ”.

This is easy to prove for any specific n :

$P_1: 1 = \frac{1}{2}(1 + 1)$ so P_1 is **TRUE**.

$P_2: 1 + 2 = 3 = \frac{2}{2}(2 + 1)$ so P_2 is **TRUE**.

$P_3: 1 + 2 + 3 = 6 = \frac{3}{2}(3 + 1)$ so P_3 is **TRUE**.



But how can we prove this for every possible n at once?

We are proving: $1 + \dots + n = \frac{n}{2}(n + 1)$.

This is very easy to quickly prove for all n using **induction**.

In an induction proof you always have to do exactly two things:

Step 1: Prove the first case P_1 . (This is called the **base case**.)

In this example the base case is absolutely trivial to see:
The sum of the first 1 positive integers is 1 which equals
the RHS for this case: $\frac{1}{2}(1 + 1)$.

Thus the base case P_1 is proved.

(We are proving: $1 + \dots + n = \frac{n}{2}(n+1)$. Continued.)

When P_n is true,
 P_{n+1} also true.

Step 2: Prove that whenever the case P_n is true then the case P_{n+1} is also true. (This is called the *induction case*.)

So in this step we are allowed to assume the n -th case P_n is true

$$1 + \dots + n = \frac{n}{2}(n+1) \quad (P_n)$$

and we must show the $(n+1)$ -st case follows from it.

The proof of P_{n+1} using P_n :

$$LHS = \overbrace{1 + \dots + n}^{\text{LHS of } P_n} + (n+1) \quad \text{(This is where we use } P_n \text{.)}$$

$$= \frac{n}{2}(n+1) + (n+1)$$

$$= (n+1) \left(\frac{n}{2} + 1 \right)$$

$$= \frac{n+1}{2} ((n+1) + 1) = RHS \text{ of } P_{n+1}$$

Step 1 and Step 2 imply
 P_n is true for every n . So
the proof is finished.



$$\begin{aligned} &= \frac{(n+1)(n+2)}{2} = \frac{n+1}{2} (n+1 + 1) \end{aligned}$$

Video 7.1.2

Determinants

We are now going to start learning about the general theory of determinants.

First comments:

You can only take a determinant of a **square** matrix. The determinant of a square matrix A is a **single real number** $\det A \in \mathbb{R}$ calculated using **all** the entries of A .

$$\det : \{\text{Square matrices}\} \rightarrow \mathbb{R}.$$

The 1×1 case

The case of a square 1×1 -matrix is trivial: $\det([c]) = c$.

The 2×2 case

Hopefully you've met the 2×2 case already, which can be explicitly defined by the formula:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

An alternative notation

Sometimes it is convenient to use a simpler notation for determinant. The following are two notations for the same thing:

$$\det \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} = \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}.$$

Why are they useful?

Actually there are many many different reasons.

We'll see in this course that some of the most important reasons are:

- It tells you *immediately* if a square matrix A has an inverse or not: If $\det A \neq 0$ then there does exist an inverse matrix A^{-1} to A . On the other hand, if $\det A = 0$ then there definitely *does not exist an inverse*.
- Moreover, in the case $\det A \neq 0$, determinant not just tells you the inverse exists but goes further and *gives you an explicit formula* for it. (This is called Cramer's rule.)
- Because $\det(AB) = \det A \det B$ determinant replaces *non-commutative* matrix products with *commutative* products in $\mathbb{R}$. This gives a simplified "picture" (the precise math word is "homomorphism") of matrix algebra inside $\mathbb{R}$.

Why are they sometimes not so useful?

Sometimes they are extremely useful, but note that there are some significant drawbacks:

- Because the fundamental definition is recursive, the time it takes to compute them grows extremely quickly with the size of the matrix. The reason is that if A is an $n \times n$ matrix then the determinant is a sum of $n!$ terms.
- It can be a powerful tool but because it is defined via a “mysterious” formula, the determinant sometimes hides the mathematical structure of what you are studying. For example sometimes it is better to understand properties of the specific matrix you are studying rather than just give it to a computer to calculate the determinant.

The fundamental definition of Determinant

Video 7.1.3

We will learn many many properties and tricks for calculating determinants, but as always in mathematics we need to start with a fundamental definition from which we derive all the other formulae.

The fundamental definition we'll use here is also called the **cofactor expansion** of the determinant. We need to build up to it with two preliminary definitions:

- The **minor matrix** M_{ij} associated to row i and column j .
- The **cofactor** C_{ij} associated to row i and column j .

Our definition is going to be **recursive**. First we define 2×2 determinants, then we define 3×3 determinants using our definition of 2×2 determinants, then we define 4×4 determinants using our definition of 3×3 determinants, and so on and so on...

Definition (Minor matrices of a matrix.)

Let A be an $n \times n$ matrix. Let i and j be integers such that $1 \leq i \leq n$ and $1 \leq j \leq n$.

Then the (i, j) -**minor matrix** of A is the $(n - 1) \times (n - 1)$ matrix denoted M_{ij} which you get when you forget the i th-row, and the j th-column.

Example:

If $A = \begin{bmatrix} 1 & 0 & -3 & 5 \\ 1 & -4 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 1 & 3 & -10 \end{bmatrix}$ then the $(2,4)$ -minor matrix M_{24} is obtained as follows:

$$\begin{bmatrix} 1 & 0 & -3 & 5 \\ 1 & -4 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 1 & 3 & -10 \end{bmatrix} \Rightarrow M_{24} = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}.$$

Definition (Cofactor of an entry in a matrix.)

Let A be an $n \times n$ matrix. Let i and j be integers such that $1 \leq i \leq n$ and $1 \leq j \leq n$. (We think of i as the row number, and j as the column number.)

Then the (i, j) -**cofactor** of A is the number

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$

Some language. I'll refer to:

- The matrix M_{ij} as the (i, j) -minor matrix.
- It's determinant $\det(M_{ij})$ is called the (i, j) -minor.

Definition (Cofactor of an entry in a matrix.)

Let A be an $n \times n$ matrix. Let i and j be integers such that $1 \leq i \leq n$ and $1 \leq j \leq n$. (We think of i as the row number, and j as the column number.)

Then the (i, j) -**cofactor** of A is the number

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$

“Hang on! You are defining cofactors so you can introduce determinants, but your definition of cofactor itself uses determinants! Isn’t this a crazy circular definition?”

Why is this still OK?

Because this determinant is of a **smaller matrix**. So it can be defined earlier in the process. The definitions are recursive, not circular.

Example:

Let’s calculate a few cofactors of this 3×3 matrix. These will need 2×2 determinants, which is fine because we defined these already.

What is C_{32} ?

The corresponding minor matrix M_{32} is

$$M_{32} = \begin{bmatrix} 1 & 1 & 4 \\ -1 & 5 & 0 \\ 0 & 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ -1 & 0 \end{bmatrix}$$

Then:

$$\begin{aligned} C_{32} &= (-1)^{3+2} \det M_{32} \\ &= (-1) * ((1)(0) - (4)(-1)) \\ &= -4 \end{aligned}$$

What is C_{13} ?

The corresponding minor matrix M_{13} is

$$M_{13} = \begin{bmatrix} 1 & 1 & 4 \\ -1 & 5 & 0 \\ 0 & 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 5 \\ 0 & 2 \end{bmatrix}$$

Then:

$$\begin{aligned} C_{13} &= (-1)^{1+3} \det M_{13} \\ &= 1 * ((-1)(2) - 5(0)) \\ &= -2 \end{aligned}$$

Definition

Let A be an $n \times n$ matrix. Let i and j be integers such that $1 \leq i \leq n$ and $1 \leq j \leq n$. (We think of i as the row number, and j as the column number.)

Then the (i, j) -**cofactor** of A is the number

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$



Note that this sign is not hard to remember when you are calculating. It starts at $(+1)$ in the top left corner of the matrix, then you change sign every time to step into the next row or column.

$$\begin{bmatrix} +1 & -1 & +1 & -1 & +1 \\ -1 & +1 & -1 & +1 & -1 \\ +1 & -1 & +1 & -1 & +1 \\ -1 & +1 & -1 & +1 & -1 \\ +1 & -1 & +1 & -1 & +1 \end{bmatrix}.$$

Definition (Determinant)

Let A be an $n \times n$ matrix.

The **determinant** of A is given by the following formula

$$\det A = a_{11}C_{11} + \cdots + a_{1n}C_{1n}$$

where:

- the (i, j) -entry of A is denoted a_{ij} as usual.
- the (i, j) -cofactor of A is denoted C_{ij} as usual.

Comment:

If we decode what the formula $\det A = a_{11}C_{11} + \cdots + a_{1n}C_{1n}$ is saying we see that the definition is really rather simple.

In words it says:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

1. Go across the **top row**.
2. At every entry a_{1j} write down $a_{1j}C_{1j}$, which is just the entry itself multiplied by its cofactor.
3. Add all of these entries up.
The result is $\det A$.

Example:

Video 7.1.4

Let's do a 3×3 determinant in detail. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix}.$$

What is $\det A$?

Calculation in detail

The determinant is a sum of three terms, one for each term in the top row.

(1,1)-entry.

(1,2)-entry.

(1,3)-entry.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix}.$$

Calculation in detail

There are three terms, one for each entry in the top row:

Term corresponding to (1,1) position:

$$(1) * \left[(-1)^{1+1} \det \begin{bmatrix} \cancel{1} & \cancel{2} & \cancel{3} \\ -1 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix} \right] = 1 * 1 * (0 * 2 - (1)(-1)) = 1$$

Term corresponding to (1,2) position:

$$(2) * \left[(-1)^{1+2} \det \begin{bmatrix} \cancel{1} & \cancel{3} \\ -1 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix} \right] = 2 * (-1) * ((-1) * 2 - (1)(3)) = 10$$

Term corresponding to (1,3) position:

$$(3) * \left[(-1)^{1+3} \det \begin{bmatrix} \cancel{1} & \cancel{2} \\ -1 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix} \right] = 3 * (1) * ((-1) * (-1) - (0)(3)) = 3$$

Thus: $\det A = 1 + 10 + 3 = 14$.

Exercise:
Video 7.1.5

For the following 4×4 matrix, write down an expression for its determinant as a combination of determinants of 3×3 matrices.

$$A = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}$$

$$a \left[(-1)^{11} \begin{vmatrix} f & g & h \\ j & k & l \\ n & o & p \end{vmatrix} \right]$$

Solution:

Writing down the cofactor expansion along the top row, according to the definition we find $\det A$ equals:

$$a \begin{vmatrix} f & g & h \\ j & k & l \\ n & o & p \end{vmatrix} - b \begin{vmatrix} e & g & h \\ i & k & l \\ m & o & p \end{vmatrix} + c \begin{vmatrix} e & f & h \\ i & j & l \\ m & n & p \end{vmatrix} - d \begin{vmatrix} e & f & g \\ i & j & k \\ m & n & o \end{vmatrix}$$

Video 7.1.6

A surprising fact: we can use any row or column for our cofactor expansion.

Earlier we wrote down the rule for calculating the determinant:

In words it says:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

1. Go across the top row.
2. At every entry a_{1j} write down $a_{1j}C_{1j}$, which is just the entry itself multiplied by its cofactor.
3. Add all of these entries up.
The result is $\det A$.

Actually we don't need to use the top row for this. We can use ANY row or column and we get exactly the same answer.

So, for example, it turns out that the following rule gives *exactly the same answer*:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

1. Go down this column.
2. At every entry a_{i2} write down $a_{i2}C_{i2}$, which is just the entry itself multiplied by its cofactor.
3. Add all of these entries up.
The result is $\det A$.

(The explanation of the proof of this is too long to explain deeply during this course, but we'll say something later about a way to understand this.)

Example:

Earlier we used the cofactor expansion along the top row to

calculate that $\det \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix} = 14.$

Let's do this again using different choices of row/column to expand along and compare our answers:

Along the first column:

$$(1) * \left[(+1) \det \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \right] + (-1) \left[(-1) \det \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \right] \\ + (3) \left[(+1) \det \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \right] = 1 + 7 + 6 = 14.$$

Example (cont.):

Earlier we used the cofactor expansion along the top row to

calculate that $\det \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix} = 14.$

Along the second row:

$$(-1) * \left[(-1) \det \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \right] + (0) \left[(+1) \det \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix} \right] \\ + (1) \left[(-1) \det \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \right] = 7 + 0 + (-1)(-7) = 14.$$

Along the third column:

$$(3) * \left[(1) \det \begin{bmatrix} -1 & 0 \\ 3 & -1 \end{bmatrix} \right] + (1) \left[(-1) \det \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \right] \\ + (2) \left[(1) \det \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \right] = 3 + (7) + (2)(2) = 14.$$

Property 2 $\rightarrow$ needs induction to prove.

$$\det \begin{pmatrix} 1 & 1 & -5 \\ 10 & 2 & -6 \\ 100 & 3 & -7 \end{pmatrix} = \det \begin{pmatrix} 1 & 10 & 100 \\ 1 & 2 & 3 \\ -5 & -6 & -7 \end{pmatrix}$$

Property 3

$$f(c_1 \vec{v}_1 + c_2 \vec{v}_2) = c_1 f(\vec{v}_1) + c_2 f(\vec{v}_2)$$

 $\rightarrow$ will obey this

Property 4

$$\det \begin{pmatrix} 1 & 2 & 3 \\ -1 & \frac{2}{2} & 1000 \\ 1 & \frac{2}{2} & 3 \end{pmatrix} = 0$$

Property 5

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 10 & 100 & 1000 \\ -1 & -7 & -7 \end{pmatrix} = (-1) \det \begin{pmatrix} 1 & 2 & 3 \\ -1 & 100 & 1000 \\ -1 & 6 & -7 \end{pmatrix}$$

Property 6

$$\det \begin{pmatrix} 2 & \pi & 42 & 3 & 7 \\ 0 & 5 & -10 & 2 & 1 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 101 \end{pmatrix} = 2 \times 5 \times 2 \times 101 = 2020$$

Video 7.2.1

6 elementary properties of determinants.

We'll start thinking about determinants by deriving a few important elementary properties:

1. If a matrix A has a row of zeroes, or a column of zeroes, then $\det A = 0$.
 $\det \begin{pmatrix} 1 & 0 & -5 \\ 10 & 0 & -6 \\ 100 & 0 & -7 \end{pmatrix} = 0$
2. Recall that A^T denotes the transpose of the matrix A . These two matrices have the same determinant: $\det A^T = \det A$.
3. Determinants are **linear functions** with respect to individual rows, and also individual columns.
respects the two algebraic operations of vectors
4. If a matrix A has a repeated row, or a repeated column, then $\det A = 0$.
5. If we obtain a matrix A' from a matrix A by swapping two rows, or swapping two columns, then $\det A' = -\det A$.
6. The determinant of a triangular matrix is just the product of the entries on the diagonal.

Elementary property #1:

A row of zeroes, or a column of zeroes, gives zero det.

Observation

Let A be an $n \times n$ matrix and assume A has a row of zeroes, or a column of zeroes.

Then:

$$\det A = 0.$$

Why?

Just do a cofactor expansion along the row or column of zeroes. Obviously every term is zero.

Video 7.2.2

Elementary property #2:**Taking a transpose doesn't affect determinant.**

Recall that A^T denotes the transpose matrix of A . Recall that this means the (i, j) -entry of A^T is the (j, i) -entry of A .

Definition (Determinant of a transpose.)

Let A be an $n \times n$ matrix.

Then:

$$\det A^T = \det A.$$

$$P_n \quad \det(A^T) = \det(A)$$

Proof:

We'll do this by induction on the dimension n of the $n \times n$ matrix A .

The base case will be $n = 1$. This case is trivial because in this case A^T is precisely A .

Proof (continued):

Now let's focus on the **induction step**:

What are we assuming?

That if A is a square matrix whose order (i.e. number of rows, or columns) is strictly less than n then $\det A^T = \det A$.

What are we trying to prove based on this assumption?

That if A is a square $n \times n$ matrix then in this case also

$$\det A^T = \det A.$$

Our strategy is just to compute the determinant of A^T using a cofactor expansion, and then identify every term that appears as some corresponding term in a cofactor expansion of the determinant of A .

So we'll need to somehow relate the cofactors of A^T with the cofactors of A .

Proof (continued):

Let's focus on the cofactors then. Let C_{ij} denote the (i, j) -cofactor of A , and let C_{ij}' denote the (i, j) -cofactor of A^T .

First recall the definition: C_{ij} , the (i, j) -cofactor of A , is defined to be

$$C_{ij} = (-1)^{i+j} \det M_{ij}$$

where M_{ij} is the (i, j) -minor matrix of A (the matrix you get by removing row i and column j).

Question:

Using the induction assumption, how are the C_{ij} and C_{ij}' related?

Proof (continued):

Question:

Using the induction assumption, how are the C_{ij} and C_{ij}' related?

Answer:

The key observation is that the (i, j) -minor matrix M_{ij} of A is exactly the **transpose** of the (j, i) -minor M_{ji}' of A^T . Call this observation $(\star)$.

For example, if $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, then:

- The $(1, 3)$ -minor matrix of A equals $M_{13} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 7 & 8 \end{bmatrix}$.
- This is exactly the transpose of: the $(3, 1)$ -minor matrix of A^T which equals $M_{31}' = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 5 & 8 \end{bmatrix}$.

$$M_{13} = (M_{31}')^T$$

Proof (continued):

Question:

Using the induction assumption, how are the C_{ij} and C_{ij}' related?

Answer (cont):

The key observation is that the (i, j) -minor matrix M_{ij} of A is exactly the **transpose** of the (j, i) -minor matrix M_{ji}' of A^T . Call this observation $(\star)$.

Thus:

$$\begin{aligned}
 C_{ij} &= (-1)^{i+j} \det M_{ij} && \text{Definition of cofactor.} \\
 &= (-1)^{i+j} \det(M_{ji}')^T && \text{By the observation } (\star). \\
 &= (-1)^{i+j} \det(M_{ji}') && \text{By the induction assumption} \\
 & && \text{which we can use here because} \\
 & && \text{the minor matrix is a smaller matrix.} \\
 &= C_{ji}'
 \end{aligned}$$

Call the deduction $C_{ij} = C_{ji}'$ by $(\star\star)$.

Proof (continued):

Now we know that $C_{ij} = C_{ji}'$ $(\star\star)$ we can quickly finish the induction step (which finishes the proof).

In the calculation below let:

- a_{ij} denote the (i, j) entry of A .
- a'_{ij} denote the (i, j) entry of A^T .

Thus $a'_{ij} = a_{ji}$.

$$\begin{aligned}
 \det A^T &= a_{11}'C_{11}' + \cdots + a_{1n}'C_{1n}' && \left(\begin{array}{c} \text{Cofactor expansion} \\ \text{along row 1.} \end{array} \right) \\
 &= a_{11}C_{11}' + \cdots + a_{n1}C_{n1}' && \left(\text{Because } a_{ij}' = a_{ji}. \right) \\
 &= a_{11}C_{11} + \cdots + a_{n1}C_{n1} && \left(\text{By } (\star\star). \right) \\
 &= \det A. && \text{Cofactor expansion of } A \text{ along column 1}
 \end{aligned}$$

The final equation is just the cofactor expansion of the determinant of A along the first column.



Example:

Let's check this for a 3×3 determinant. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$.

Doing a cofactor expansion along the top row we get:

$$\det A = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Now consider the transpose: $A^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$.

Doing a cofactor expansion of the determinant down the first column write down exactly the same expression:

$$\det A^T = a(ei - fh) - b(di - fg) + c(dh - eg).$$



Elementary property #3:

Video 7.2.3

Determinants are linear functions in rows and columns.

The idea of a “linear function” is something you’ll study closely and more generally in Linear Algebra II.

Here we’ll explain what we mean with some specific formulae.

Consider an $n \times n$ matrix A .

For convenience we’ll name its rows R_1 through to R_n . (So we can think of these as $1 \times n$ matrices.)

In this discussion we are going to focus on its i -th row R_i .

So here is what A looks like:

$$A = \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix}.$$

Now assume that the i -th row R_i of A is some **linear combination** of some other $1 \times n$ matrices S and T with coefficients $x_1, x_2 \in \mathbb{R}$.

$$R_i = x_1 S + x_2 T.$$

In other words A looks like:

$$A = \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ \leftarrow & \vdots & \rightarrow \\ \leftarrow & x_1 S + x_2 T & \rightarrow \\ \leftarrow & \vdots & \rightarrow \\ \leftarrow & R_n & \rightarrow \end{bmatrix}.$$

$$\begin{aligned} & 4 \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} + 8 \begin{bmatrix} -1/2 & -5 & 0 \end{bmatrix} \\ & = \begin{bmatrix} 4 & 8 & 12 \end{bmatrix} + \begin{bmatrix} -4 & -40 & 0 \end{bmatrix} \\ & = \begin{bmatrix} 0 & -32 & 12 \end{bmatrix} \\ & \det \begin{bmatrix} 1 & 0 & 1 \\ 0 & -32 & 12 \\ 1 & -1 & 0 \end{bmatrix} = 4 \det \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & -1 & 0 \end{bmatrix} + 8 \det \begin{bmatrix} 1 & 0 & 1 \\ -1/2 & -5 & 0 \\ 1 & -1 & 0 \end{bmatrix} \\ & = 4 [3 + (-3)] + 8 [0 + 1 \cdot (\frac{1}{2} + 5)] \\ & = 44 = 44 \end{aligned}$$

Theorem (Linearity of determinant in a specific row.)

Let A be an $n \times n$ matrix as discussed above. Then

Then $\det A$ equals:

$$= x_1 \det \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ \leftarrow & \vdots & \rightarrow \\ \leftarrow & S & \rightarrow \\ \leftarrow & \vdots & \rightarrow \\ \leftarrow & R_n & \rightarrow \end{bmatrix} + x_2 \det \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ \leftarrow & \vdots & \rightarrow \\ \leftarrow & T & \rightarrow \\ \leftarrow & \vdots & \rightarrow \\ \leftarrow & R_n & \rightarrow \end{bmatrix}.$$

Example:

Consider the following 3×3 matrix:

$$\begin{bmatrix} a & b & c \\ x_1 d + x_2 p & x_1 e + x_2 q & x_1 f + x_2 r \\ g & h & i \end{bmatrix}.$$

Note that the second row is a linear combination of two other rows:

$$x_1 [d \quad e \quad f] + x_2 [p \quad q \quad r].$$

Now calculate the determinant:

$$\begin{aligned} & (x_1 d + x_2 p)(bi - ch) - (x_1 e + x_2 q)(ai - cg) + (x_1 f + x_2 r)(ah - bg) \\ = & x_1 (d(bi - ch) - e(ai - cg) + f(ah - bg)) \\ & + x_2 (p(bi - ch) - q(ai - cg) + r(ah - bg)) \\ = & x_1 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + x_2 \begin{vmatrix} a & b & c \\ p & q & r \\ g & h & i \end{vmatrix}. \end{aligned}$$

Comment:

Note that because we know that determinant is not affected by taking a transpose, we can immediately deduce a similar property with respect to columns:

Theorem (Linearity of determinant in a specific column.)

Determinant satisfies the following identity:

$$\det \left(\begin{bmatrix} \uparrow & & \uparrow & & \uparrow \\ C_1 & \cdots & x_1 S + x_2 T & \cdots & C_n \\ \downarrow & & \downarrow & & \downarrow \end{bmatrix} \right)$$

equals

$$x_1 \det \left(\begin{bmatrix} \uparrow & & \uparrow & & \uparrow \\ C_1 & \cdots & S & \cdots & C_n \\ \downarrow & & \downarrow & & \downarrow \end{bmatrix} \right) + x_2 \det \left(\begin{bmatrix} \uparrow & & \uparrow & & \uparrow \\ C_1 & \cdots & T & \cdots & C_n \\ \downarrow & & \downarrow & & \downarrow \end{bmatrix} \right).$$

Proof (of the identity for rows):

The proof is very simple. We'll just do a cofactor expansion of the determinant along row i and then just expand and regroup the brackets that appear.

First we need to name some entries for the discussion.

- Let $S = [s_1 \ \cdots \ s_n]$.
- Let $T = [t_1 \ \cdots \ t_n]$.
- Let $R_i = [r_1 \ \cdots \ r_n]$.

Thus because $R_i = c_1 S + c_2 T$ then $r_j = c_1 s_j + c_2 t_j$.

Proof (cont.):

Now we just calculate $\det A$ by doing a cofactor expansion along row i :

$$\begin{aligned}\det A &= r_1 C_{i1} + \cdots + r_n C_{in} \\ &= (c_1 s_1 + c_2 t_1) C_{i1} + \cdots + (c_1 s_n + c_2 t_n) C_{in} \\ &= c_1 (s_1 C_{i1} + \cdots s_n C_{in}) + c_2 (t_1 C_{i1} + \cdots + t_n C_{in})\end{aligned}$$

The first term is exactly c_1 times the determinant of the matrix you get by replacing row i with S .

And the second term is exactly c_2 times the determinant of the matrix you get by replacing row i with T .



Video 7.2.4

Brief Abstract Digression:

What do we mean by the term “Linear function” ?

What is the significance?

- Speaking very generally, Mathematics studies examples of mathematical structures and functions between them that *respect the structures*.
- In the case of Linear Algebra the underlying mathematical structure is something called a **Vector space**.

In LAlI you’ll learn a **Vector space**:

- Is a set V
- With an operation of addition $+$ defined.
- And an operation of scalar multiplication defined.
- Satisfying a list of rules.

Brief Abstract Digression (continued):

Linear Algebra studies:

- Vector spaces.
- Maps between vector spaces respecting the structure.

What is the structure present in a vector space?

- The addition $+$
- The scalar multiplication.

Definition:

A **Linear Transformation** between vector spaces V_1 and V_2 is a function $T: V_1 \rightarrow V_2$ that “respects the vector space structure” which here means that for all $\mathbf{v}, \mathbf{w} \in V_1$ and $r \in \mathbb{R}$ then

1. $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$.
2. $T(r\mathbf{v}) = rT(\mathbf{v})$.

Brief Digression (continued):

From this abstract point of view **Elementary property #3** is saying something very simple and natural:

“If we substitute fixed vectors into all rows of a determinant but one, then the rule that says put a vector from $\mathbb{R}^n$ into the remaining empty row then take the determinant gives us a function

$$\mathbb{R}^n \rightarrow \mathbb{R}.$$

This function is a linear transformation from the vector space $\mathbb{R}^n$ to the vector space $\mathbb{R}$.”

(Note: The special case of a linear transformation $V \rightarrow \mathbb{R}$ when the codomain is $\mathbb{R}$ is called a linear **function**, not just a linear transformation.)

Video 7.2.5

Elementary property #4:***Repeated row or column means determinant is zero.*****Theorem**

Let A be an $n \times n$ matrix. If A has a repeated row, or a repeated column, then $\det A = 0$.

Example:

It is easy to check, for example, that

$$\begin{vmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{vmatrix} = 0.$$

Proof

To begin, note that we just have to explain the property for the case of a repeated row. The case of a repeated column then follows for free. The reason is that if a matrix A has a repeated column, then its determinant equals the determinant of its transpose, which will have a repeated row, so $\det A = \det A^T = 0$.

We'll establish this property by **induction** on n the dimension of the $n \times n$ matrix A .

The base of the induction:

The base case of the induction will be the case $n = 2$.

This case is easy. If $n = 2$ then we know exactly what every matrix with a repeated row looks like, because there are only two rows.

So can just calculate directly: $\begin{vmatrix} a & b \\ a & b \end{vmatrix} = ab - ba = 0$.

base case is not always 1.

simplest matrix with repeated rows. if $n=1$, cannot have repeated row.

Proof (continued)

Now we focus on the induction step. Let be n be an integer $n > 2$.

What are we assuming?

That the determinant of an $m \times m$ matrix with a repeated row whose dimension m is strictly less than n is zero.

What do we have to prove based on this assumption?

That the determinant of an $n \times n$ matrix with a repeated row is also zero.

Let A be an $n \times n$ matrix. Assume row i and row j are the same.

Do a cofactor expansion of $\det A$ along any choice of row k other than i or j :

$$\det A = a_{k1}C_{k1} + \cdots + a_{kn}C_{kn}.$$

↓
they will be cancelled out when we find the det. so no more identical rows

Proof (continued)

Do a cofactor expansion of $\det A$ along a choice of row k other than i or j :

$$\det A = a_{k1}C_{k1} + \cdots + a_{kn}C_{kn}.$$

Note that each of these cofactors C_{kl} is zero so $\det A = 0$ as required. **Why are these cofactors all zero?**

The answer:

Each cofactor is of the form (± 1) multiplied by $\det M_{kl}$, where M_{kl} is the matrix obtained from A by deleting row k and column l .

Because row k was chosen different to the repeated rows row i and row j , this matrix still has a repeated row, but is one dimension smaller.

Because it is smaller we can apply the induction assumption and deduce: $C_{kl} = (-1)^{k+l} \det M_{kl} = 0$.



Video 7.2.6

Elementary property #5:**Effect on determinant of a row swap, or column swap.****Theorem**

Let A be an $n \times n$ matrix. Let A' be a matrix obtained from A by swapping two rows, or by swapping two columns.

Then:

$$\det A' = -\det A.$$

For example, you can easily check that

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 2 & 0 & -1 \end{vmatrix} = 1(-1 - 0) - 2(-1 - 2) + 3(0 - 2) = -1$$

while the result of swapping column 2 and 3:

$$\begin{vmatrix} 1 & 3 & 2 \\ 1 & 1 & 1 \\ 2 & -1 & 0 \end{vmatrix} = 1(0 + 1) - 3(0 - 2) + 2(-1 - 2) = 1.$$

Proof

Again, note that we just need to explain why the property is true **for a row swap**. The case of a column swap then follows for free by taking the transpose of the matrix.

So consider a matrix A as follows and a matrix A' obtained by swapping rows i and j :

$$A = \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i & \rightarrow \\ & \vdots & \\ \leftarrow & R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix}, \quad A' = \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_i & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix}.$$

• Use linearity in rows
 • repeated rows $\Rightarrow$ det is 0.

How can we relate the two determinants to each other?

Proof (continued)

The trick is super simple.

Replace both row i and row j by the sum of the two rows $R_i + R_j$.

The resulting matrix M looks like:

$$\begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix}.$$

We'll simply calculate the determinant of this matrix in two different ways, using properties of the determinant we have already proved.

Proof (continued)

We are studying:

$$M = \begin{bmatrix} \leftarrow & R_1 & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_i + R_j & \rightarrow \\ & \vdots & \\ \leftarrow & R_n & \rightarrow \end{bmatrix}.$$

Calculation #1:

Note that M has a repeated row. Thus $\det M = 0$.

Video 7.2.7

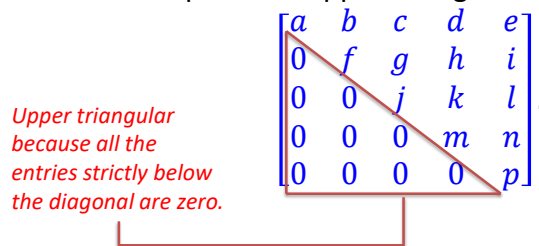
Elementary property #6:**Determinant of a triangular matrix.**

Recall that a **triangular** matrix is a square matrix such that either

- all the entries below the diagonal are zero (this case is called **upper triangular**)
- or all the entries above the diagonal are zero (this case is called **lower triangular**).

Here is an example of an upper triangular matrix:

Upper triangular
because all the
entries strictly below
the diagonal are zero.

$$\begin{bmatrix} a & b & c & d & e \\ 0 & f & g & h & i \\ 0 & 0 & j & k & l \\ 0 & 0 & 0 & m & n \\ 0 & 0 & 0 & 0 & p \end{bmatrix}.$$


Question:

If you take the transpose of an upper triangular matrix, what do you get?

Answer:

A lower triangular matrix with the **same diagonal**.

For example:

$$\begin{bmatrix} a & b & c & d & e \\ 0 & f & g & h & i \\ 0 & 0 & j & k & l \\ 0 & 0 & 0 & m & n \\ 0 & 0 & 0 & 0 & p \end{bmatrix}^T = \begin{bmatrix} a & 0 & 0 & 0 & 0 \\ b & f & 0 & 0 & 0 \\ c & g & j & 0 & 0 \\ d & h & k & m & 0 \\ e & i & l & n & p \end{bmatrix}$$

Theorem

Let A be a triangular $n \times n$ matrix.

Then $\det A$ is the product of the diagonal entries.

Example

$$\det \begin{bmatrix} a & b & c & d & e \\ 0 & f & g & h & i \\ 0 & 0 & j & k & l \\ 0 & 0 & 0 & m & n \\ 0 & 0 & 0 & 0 & p \end{bmatrix} = afjmp.$$

Example

$$\det \begin{bmatrix} 2 & & 0 & 0 \\ 10000 & & 3 & 0 \\ & & & \\ 1 & 20000 & & -1 \end{bmatrix} = -6.$$

Theorem

Let A be a triangular $n \times n$ matrix.

Then $\det A$ is the product of the diagonal entries.

Caution:

This only works when your matrix is *triangular*: all zeros strictly below the diagonal (or all zeros strictly above the diagonal).

Many students forget this in exams and just assume every determinant is the product of its diagonal.

Question:

What is the determinant of the following matrix:

$$\begin{bmatrix} 216 & -1 & 1 & -1 & 1 \\ 0 & -42 & 1 & -1 & -1 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 52 & -1 \\ 0 & 0 & 0 & 0 & -291 \end{bmatrix} ?$$

This is an upper triangular matrix, so its determinant is the product of diagonal entries. One of those diagonal entries is 0, so the determinant is 0.

Proof:

We just need to explain the case of upper triangular matrices.

Why?

The determinant of the transpose of a matrix equals the original determinant. So:

- The determinant of a lower triangular matrix A
- Equals the determinant of its transpose A^T
- Equals the product of the diagonal entries of A^T
- Equals the product of the diagonal entries of A .

Proof (continued):

We will explain the case of upper triangular matrices by induction on n , the dimension of the $n \times n$ matrix A .

The base case:

For the case $n = 1$ the fact is obviously true. The determinant equals the only entry in the matrix which equals the product of diagonal entries.

Proof (continued):***The induction step:******What are we assuming?***

That for every upper triangular matrix of dimension strictly less than n the determinant equals the product of the diagonal entries.

What must we explain based on this assumption?

That this fact is also true for $n \times n$ upper triangular matrices.

So let A be an upper triangular $n \times n$ matrix. It looks like the following:

$$\begin{bmatrix} a_{11} & * & * & * \\ 0 & a_{22} & * & * \\ \vdots & \vdots & \ddots & * \\ 0 & 0 & 0 & a_{nn} \end{bmatrix}.$$

Proof (continued):

Calculate $\det A$ by doing a cofactor expansion down the first column:

$$\det A = a_{11} \det \begin{bmatrix} * & * & * & * \\ 0 & a_{22} & * & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & a_{nn} \end{bmatrix} + 0 + \dots + 0$$

$$= a_{11} * (a_{22} \dots a_{nn}),$$

where to get the second equality we have used the induction assumption. We can use the induction assumption here because the matrix

$$\begin{bmatrix} a_{22} & * & * \\ \vdots & \ddots & \vdots \\ 0 & 0 & a_{nn} \end{bmatrix}$$

is again an upper triangular matrix and is strictly smaller than A .

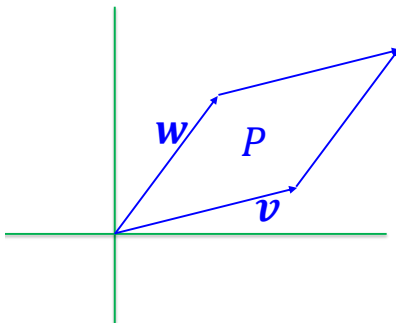
□

A geometric interpretation of determinant:

Video 7.2.8

Two dimensions:

Take two vectors $v, w \in \mathbb{R}^2$:



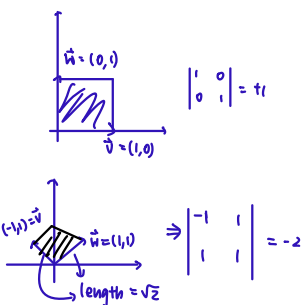
Challenging exercise:

Prove that

$$\det \begin{pmatrix} \uparrow & \uparrow \\ v & w \\ \downarrow & \downarrow \end{pmatrix} = \pm \text{Area}(P)$$

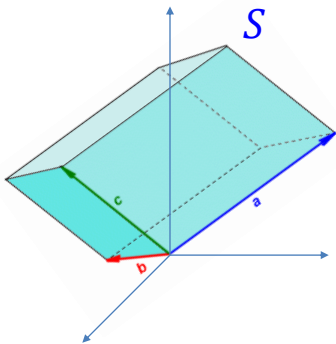
where you use the right hand rule to determine the sign. (When your fingers of your right hand curve from v to w if your thumb sticks out of the page then the sign is $+$. Otherwise $-$.)

* right hand grip rule



Three dimensions:

Take three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$.



Fact: (Tricky to prove, but try if interested!)

It is a fact that

$$\det \begin{pmatrix} \uparrow & \uparrow & \uparrow \\ \mathbf{a} & \mathbf{b} & \mathbf{c} \\ \downarrow & \downarrow & \downarrow \end{pmatrix} = \pm \text{Vol}(S)$$

where you use the right hand rule to determine the sign. (When your fingers of your right hand curve from $\mathbf{a}$ to $\mathbf{b}$ if your thumb sticks in the direction of $\mathbf{c}$ then the sign is $+$. Otherwise $-$.)

Higher dimensions:

In dimensions higher than 3, we use determinant to *define* a concept of volume. Please take MH4600 Differential Geometry if you are interested to understand this!

Video 7.2.9

An alternative definition of determinant based on permutations.

Sometimes the determinant is easier to understand using a different definition that is more *"balanced"*. (By this I mean the definition doesn't start by choosing a row or column.)

To introduce this definition of determinant for an $n \times n$ -matrix $\mathbf{A}$ we first need to meet the concept of a *permutation* of the numbers $\{1, \dots, n\}$.

A **permutation** π of the numbers $\{1, \dots, n\}$ is a function from this set to itself $\pi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ which is one-to-one and onto. In other words, a bijection of this set to itself.

You can think of a bijection like this as just a **rearrangement** of the ordered list of numbers $1 \dots n$ to the ordered list $\pi(1) \dots \pi(n)$.

For example, there are 24 permutations of the set $\{1, 2, 3, 4\}$ corresponding to the rearrangements:

1234	2134	3124	4123
1243	2143	3142	4132
1324	2314	3214	4213
1342	2341	3241	4231
1423	2413	3412	4312
1432	2431	3421	4321

To every permutation π we can associate a sign, $\text{sgn}(\pi) \in \{+1, -1\}$.

The definition is: $\text{sgn}(\pi) = (-1)^{\text{swaps}(\pi)}$, where $\text{swaps}(\pi)$ is the number of swaps of adjacent numbers you need to do to rearrange the numbers according to the permutation.

For example: the permutation corresponding to the rearrangement 3142 can be obtained from 1234 via **three** adjacent swaps:

$1234 \rightarrow 1324 \rightarrow 3124 \rightarrow 3142$

Thus the sign of this permutation is $\text{sgn}(\pi) = (-1)^3 = (-1)$.

Alternative "balanced" definition of Determinant

Let A be an $n \times n$ matrix.

As usual let a_{ij} denote the entry of A in row i and column j .

Then the determinant of A can be defined by the formula:

$$\det A = \sum_{\substack{\text{all} \\ \text{permutations} \\ \pi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}}} \text{sgn}(\pi) a_{1\pi(1)} \dots a_{n\pi(n)}.$$

→ for the sign

$\sum_{\text{perms } \pi} \text{sgn}(\pi) a_{1\pi(1)} \dots a_{n\pi(n)}$

Each row only 1, each column only 1.

$a_{11} a_{22} a_{33} a_{44}$

1 2 3 4
3 2 4 1

→ must be adjacent swaps
1234 → 1324 → 3124 → 3214 → 3241 there's 4 adjacent swaps
∴ $(-1)^4$

Example:

Let's use this definition to calculate the determinant of the

general 3×3 matrix: $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$

There will be one term for each of the 6 permutations

$$\pi: \{1, 2, 3\} \rightarrow \{1, 2, 3\}.$$

Let's calculate the 6 terms via a table:

Permutation	Sign	Product
123	+1	$a_{11}a_{22}a_{33}$
132	-1	$a_{11}a_{23}a_{32}$
213	-1	$a_{12}a_{21}a_{33}$
231	+1	$a_{12}a_{23}a_{31}$
312	+1	$a_{13}a_{21}a_{32}$
321	-1	$a_{13}a_{22}a_{31}$

→ sum of the 6 terms

Thus $\det A$ equals:

$$a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}.$$

123 → 132 → 312 → 321
∴ $(-1)^3 a_{13}a_{22}a_{31}$

Example (cont.):

We just calculated:

$$a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}.$$

Let's check against the result of a cofactor expansion along the top row. The

determinant $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ equals:

$$= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

These are equal, as expected.

How to match this definition with cofactor expansions?

The different cofactor expansions we met already arise from this when you organize the partitions into different subsets in different ways.

Exercise:

Re-organise the sum

as

$$\det A = \sum_{\substack{\text{all} \\ \text{permutations} \\ \pi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}}} \text{sgn}(\pi) a_{1\pi(1)} \dots a_{n\pi(n)}$$

$$\det A = \sum_{m=1}^n a_{1m} \sum_{\substack{\text{all} \\ \text{permutations} \\ \pi: \{1, \dots, n\} \rightarrow \{1, \dots, n\} \\ \text{such that } \pi(1)=m}} \text{sgn}(\pi) a_{2\pi(2)} \dots a_{n\pi(n)}.$$

entries in the top row

Handwritten notes:

- Subset of permutations where 1 is in first position
- Subset of permutations where 2 is in first position
- Subset of permutations where 3 is in first position

Handwritten examples:

	1 2 3	1 3 2
Subset of permutations where 1 is in first position	1 2 3	1 3 2
Subset of permutations where 2 is in first position	2 1 3	2 3 1
Subset of permutations where 3 is in first position	3 1 2	3 2 1

Check that this expression is precisely what you get when you do a cofactor expansion along the top row.

Challenge:

How can you reorganise the sum

$$\det A = \sum_{\substack{\text{all} \\ \text{permutations} \\ \pi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}}} \text{sgn}(\pi) a_{1\pi(1)} \dots a_{n\pi(n)}$$

to obtain all the other possible cofactor expansions of the determinant along other rows/columns?

Prove it!

Complexity of calculation

This permutation formula for the determinant lets us understand the “complexity” of the calculation better.

(“Complexity” is a technical term from computer science meaning how the duration of the computation grows as the size of the input grows.)

Here is the formula again:

$$\det A = \sum_{\substack{\text{all} \\ \text{permutations} \\ \pi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}}} \text{sgn}(\pi) a_{1\pi(1)} \dots a_{n\pi(n)}.$$

How many terms are there in this sum when A is $n \times n$?

Answer: There are $n!$ terms.

We just noted the permutation formula for determinant has $n!$ terms.

The factorial function $n!$ grows extraordinarily quickly.

Asymptotic

Stirling's formula gives a way to get a sense of this growth rate:

For large n

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

Estimate for number of atoms in Milky way: $\sim 10^{67}$

n	$n!$
0	1
1	1
2	2
3	6
4	24
5	120
6	720
7	5 040
8	40 320
9	362 880
10	3 628 800
11	39 916 800
12	479 001 600
13	6 227 020 800
14	87 178 291 200
15	1 307 674 368 000
16	20 922 789 888 000
17	355 687 428 096 000
18	6 402 373 705 728 000
19	121 645 100 408 832 000
20	2 432 902 008 176 640 000
25	$1.551\,121\,004 \times 10^{25}$
50	$3.041\,409\,320 \times 10^{64}$
70	$1.197\,857\,167 \times 10^{100}$